[Assignment 5] Automated Testing with GitHub Actions 100 Points Possible

11/22/2026

Attempt 1

In Progress

NEXT UP: Submit Assignment

Add Comment

Unlimited Attempts Allowed

10/21/2026

Details

In this assignment, you will extend your CI workflow from Assignment 4 by adding **automated tests** to your COLORS web application. These tests will run automatically using GitHub Actions on every push and pull request.

This assignment introduces the concept of **testing as part of CI/CD pipelines**, a critical practice in modern software engineering.

Learning Outcomes

By completing this assignment, you will:

- Understand the role of testing in CI/CD pipelines
- Write basic unit or integration tests
- Integrate test execution into GitHub Actions
- Interpret and debug test failures in CI

Task Overview

- Step 1: You will create a new branch
- Step 2: Add a testing framework (you are recommended to use `jest`)
- Step 3: You must have ***at least one front-end unit test*** (e.g., validate function that adds or formats colors, test form validation logic) and ***at least one integration test*** (e.g., test API endpoint response, validate JSON structure)
- Step 4: Update GitHub Actions workflow
- Step 5: Debug and ensure success

Submission Instructions

Submit the URL for the final commit ***after*** completing the assignment. This will look like **https://github.com/username/repository/commit/abcdef123456789 (https://github.com/username/repository/commit/abcdef123456789)** .

This commit must include: test files, updated GitHub Actions workflow, passing CI pipeline (lint + tests).

Important Notes

- Do not submit screenshots of test reports
- The GitHub Actions tab is the source of truth for grading
- Ensure your repository is public
- Node.js can/must be used for setting up testing tools, not for running your application (Why? LAMP project!!!)

Enter Web URL

https://

(https://webcourses.ucf.edu/courses/1520889/modules/items/21862861)

Assignment

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